DBMS Task - Sales and Customer Analytics System

Snapdeal_e-commerce_project

OBJECTIVE

1. Apply COUNT(), SUM(), AVG(), MIN(), MAX().
2. Generate total sales reports.
3. Find top customers based on purchase amount.
4. Identify best-selling products.
5. Perform category-wise sales analysis.

Objective 1: Apply COUNT(), SUM(), AVG(), MIN(), MAX()

```
SQL> SELECT
  2      COUNT(*) AS TOTAL_PRODUCTS,
  3      SUM(PRICE) AS TOTAL_PRICE,
  4      AVG(PRICE) AS AVERAGE_PRICE,
  5      MIN(PRICE) AS MIN_PRICE,
  6      MAX(PRICE) AS MAX_PRICE
  7  FROM PRODUCT;

TOTAL_PRODUCTS TOTAL_PRICE AVERAGE_PRICE  MIN_PRICE  MAX_PRICE
-------------- ----------- ------------- ---------- ----------
             4       72500         18125       1000      50000
```

Objective 2: Generate Total Sales Report

```
             4       72500         18125       1000      50000

SQL> SELECT
  2      COUNT(*) AS TOTAL_ORDERS,
  3      SUM(TOTAL_AMOUNT) AS TOTAL_SALES,
  4      AVG(TOTAL_AMOUNT) AS AVERAGE_SALES,
  5      MIN(TOTAL_AMOUNT) AS MIN_SALE,
  6      MAX(TOTAL_AMOUNT) AS MAX_SALE
  7  FROM SALES;

TOTAL_ORDERS TOTAL_SALES AVERAGE_SALES   MIN_SALE   MAX_SALE
------------ ----------- ------------- ---------- ----------
          12      209500    17458.3333       2000      52000

SQL>
```

Objective 3: Find Top Customers Based on Purchase Amount

```
SQL> SELECT
  2      C.CUSTOMER_ID,
  3      C.CUSTOMER_NAME,
  4      SUM(S.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  5  FROM CUSTOMER C, SALES S
  6  WHERE C.CUSTOMER_ID = S.CUSTOMER_ID
  7  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  8  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------
        101
Rahul
         52000

        106
Anjali
         50000

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------

        104
Sneha
         40000
```

Objective 4: Identify Best-Selling Products

```
SQL> SELECT
  2      P.PRODUCT_ID,
  3      P.PRODUCT_NAME,
  4      SUM(SI.QUANTITY) AS TOTAL_QUANTITY_SOLD
  5  FROM PRODUCT P, SALE_ITEMS SI
  6  WHERE P.PRODUCT_ID = SI.PRODUCT_ID
  7  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  8  ORDER BY TOTAL_QUANTITY_SOLD DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
TOTAL_QUANTITY_SOLD
-------------------
         4
Keyboard
                  6

         3
Headphones
                  5
```

Objective 5: Perform Category-Wise Sales Analysis

```
SQL> SELECT
  2      C.CATEGORY_NAME,
  3      SUM(SI.QUANTITY) AS TOTAL_QUANTITY,
  4      SUM(SI.QUANTITY * SI.UNIT_PRICE) AS TOTAL_SALES
  5  FROM CATEGORY C, PRODUCT P, SALE_ITEMS SI
  6  WHERE C.CATEGORY_ID = P.CATEGORY_ID
  7  AND P.PRODUCT_ID = SI.PRODUCT_ID
  8  GROUP BY C.CATEGORY_NAME
  9  ORDER BY TOTAL_SALES DESC;

CATEGORY_NAME                                      TOTAL_QUANTITY TOTAL_SALES
-------------------------------------------------- -------------- -----------
Footwear                                                        4      200000
Books                                                           3       42000
Beauty                                                          5        7500
Grocery                                                         6        6000
```

Conclusion:

Thus, the Sales and Customer Analytics System was successfully implemented using SQL. Aggregate functions such as COUNT(), SUM(), AVG(), MIN(), and MAX() were applied to analyze sales data. Total sales reports were generated, top customers were identified, best-

selling products were found, and category-wise sales analysis was performed. This system helps in understanding sales performance and making better business decisions.